English Speaking Assessment and Pronunciation Error Detection



A machine-learning support tool for English learners

GROUP MEMBERS

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Problem & User

Machine learning that supports a teacher or a learner — the course's educational-support theme.

THE PROBLEM

Learners can barely hear their own pronunciation errors, and individual teacher feedback does not scale to daily practice.

WHO IT SUPPORTS

English learners practising speaking — and teachers checking where each learner needs help.

WHAT IT INFORMS

What to practise next: pronunciation accuracy, fluency and prosody — down to the words and phonemes that need work.

Human judgement stays in charge — the tool only suggests what to review; every final decision about a learner stays with the learner or the teacher.

WHAT THE TOOL IS NOT

- Not an automatic grader
- Not an AI teacher
- Not a replacement for the teacher

Proposed Solution

From one learner recording to review-ready feedback — a support tool, not a grader.

1 · INPUT



A recording of the learner reading an English sentence

2 · ML MODELS — PLANNED

- Family 1 — acoustic features + SVM / Random Forest
- Family 2 — Mel-spectrogram + CNN

3 · SUGGESTED OUTPUTS

- Speaking assessment — accuracy · fluency · prosody · total scores
- Pronunciation errors — word- and phoneme-level

4 · HUMAN REVIEW

The learner or teacher checks every output and decides what to act on

Decision support only — every score and every flagged error is a suggestion for the learner or teacher to check; nothing is graded automatically.

Dataset — speechocean762

Data route: public benchmark · **huggingface.co/datasets/mispeech/speechocean762**

5,000
sentence recordings

250
non-native English speakers

5
expert annotators per sentence

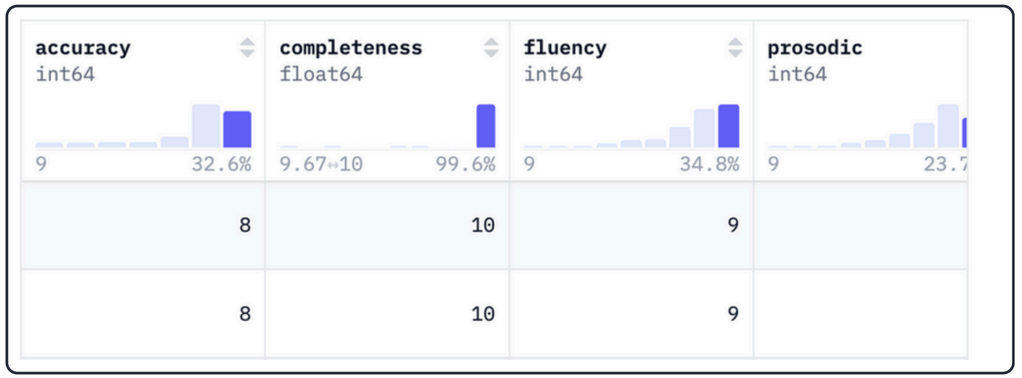
ANNOTATED AT THREE LEVELS

SENTENCE accuracy · fluency · prosody · completeness · total score

WORD pronunciation score for every word

PHONEME quality score per phoneme; mispronunciation tags for low scores

Also included: audio + reference transcripts · children and adults · built for pronunciation assessment / computer-assisted language learning.



accuracy int64	completeness float64	fluency int64	prosodic int64
9 … 32.6%	9.67…10 … 99.6%	9 … 34.8%	9 … 23.7
8	10	9	
8	10	9	

Live Dataset Viewer on the Hugging Face dataset page

Licence: CC BY 4.0 (OpenSLR / official documentation)

Machine-Learning Task

The same recordings become two supervised problems with defined inputs and outputs.

TASK A · SPEAKING ASSESSMENT

REGRESSION

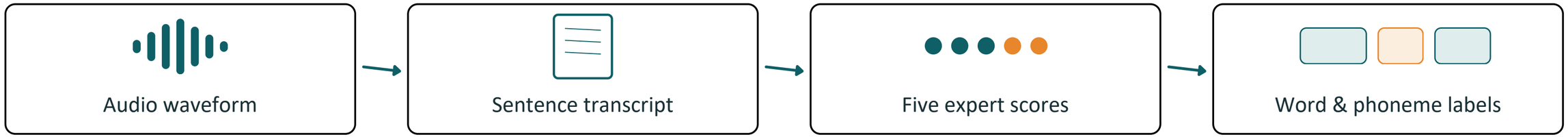
- Input — a full English utterance (audio + transcript)
- Output — predicted accuracy, fluency, prosody, total
- Targets are numerical — formulated as regression
- One prediction per recording

TASK B · ERROR DETECTION

CLASSIFICATION

- Input — audio aligned to one word or phoneme
- Output — correct vs. pronunciation error
- Two discrete classes — formulated as classification
- One prediction per word / phoneme

ONE TRAINING EXAMPLE — WHAT THE MODEL SEES



Labels are subjective human judgements — we will train on the mean of the five expert scores. Error labels are the minority class (see evaluation plan).

Model Plan — Two Model Families

Families covered in this course, trained on the same data and compared on the same splits.

FAMILY 1

Classical machine learning

ENGINEERED FEATURES
MFCC · pitch · energy · speech-duration

→

SVM / Random Forest

Interpretable and computationally simple; each feature maps to a known phonetic cue — an explainable reference point.

FAMILY 2

Neural network

INPUT REPRESENTATION
Mel-spectrogram of the utterance

→

Convolutional neural network (CNN)

Learns acoustic patterns directly from the spectrogram — can capture structure that hand-crafted features miss.

Why compare them — do learned representations improve on manually engineered acoustic features? The classical family is our interpretable reference; both must beat the named baselines.

Evaluation Plan

Speaker-independent splits, task-appropriate metrics, and named baselines.

SPLIT — SPEAKER-INDEPENDENT

TRAIN	VALIDATION	TEST
70%	15%	15%
of speakers	of speakers	of speakers

A speaker appears in exactly one split — no speaker leakage between training and test.

BASELINE — WHAT OUR MODELS MUST BEAT

Regression: predict the mean training-set score

Classification: predict the majority class (correct pronunciation)

METRICS

TASK A · REGRESSION

MAE · RMSE · Pearson correlation (where appropriate)

TASK B · CLASSIFICATION

Precision · Recall · F1 · confusion matrix

WHY ACCURACY ALONE IS NOT ENOUGH

Correct pronunciations will be the majority class: a model that always answers “correct” looks accurate while finding no errors. Precision, recall and F1 show whether it actually detects errors.

Planned Error Analysis, Risks & Contribution

What we will examine once models are trained — and the honest limits of the plan.

ERROR ANALYSIS — PLANNED

- Which phonemes are misclassified most often
- Children vs adults, and across proficiency levels
- Fluency and prosody vs accuracy — which is harder to predict
- Recordings where our prediction differs most from the human score

RISKS & LIMITATIONS

- Mandarin-L1 data — performance may not transfer to Vietnamese learners
- Human scoring is subjective — five experts reduce, not remove, this
- Microphone quality, noise and speaker characteristics affect predictions
- Pronunciation quality is not the same as overall speaking proficiency

Expected contribution: suggested accuracy, fluency, prosody and total scores, plus word- and phoneme-level errors to review — built on licensed, expert-annotated public data.

What stays human: every output is a suggestion to review; the learner or teacher decides what to act on. It is not an automatic examiner.